CORAL run 2026-10-09_004013 — results

Run summary

Item	Value
Task	cache-ext-get-scan (LevelDB GET/SCAN, cache_ext eviction-policy search)
Run start → stop (local)	00:40:13 → 01:43:33 (63.3 min)
Stop reason	max_real_attempts (cap 20, real attempts 20); triggering attempt c3bd68ae, score -0.4568
Agent	1 agent (captain-nemo), runtime claude_code, model alias claude-opus-5-5 → Bedrock us.anthropic.claude-opus-5-5
Attempts recorded	28 = 20 real + 8 --tune; coral eval invocations: 29; grader errors: 0
Attempt status (CORAL)	improved: 6, regressed: 22
Agent sessions (claude -p)	36; terminal reason: aborted_streaming 28, completed 8
LLM usage (agent)	66,643 output tokens; 14,199,873 cache-read + 324,577 cache-write + 242 input tokens; Claude Code cost estimate $6.10; API time 1,498 s
Notes / skills written	18 note files; skills present: create-notes, deep-research, organize-files, probe-table, skill-creator
Distinct code versions	23 (identical code after stripping comment-only lines)

Headline numbers

Metric	Value	Metric	Value
Noop baseline throughput (n=8)	1,224.3 ± 9.5 ops/s	Attempts above baseline mean	21 / 28
Best by score (real)	#17 61054f8b: score 0.9980, 1,257.4 ops/s	Attempts > mean+1σ / +2σ / +3σ	19 / 17 / 15
Best by throughput (all)	#11 2d5d649a (real): 1,277.7 ops/s (+4.36%), score 0.8321	Attempts below baseline mean	7 / 28
Worst by throughput	#25 df506481: 1,070.5 ops/s (-12.56%)	Score range (min … max)	-0.9040 … 0.9983
Throughput, all attempts: mean ± sd	1,228.8 ± 57.2 ops/s	Throughput median (p10 … p90)	1,254.8 (1,141.6 … 1,273.3)
Throughput range (max − min)	207.1 ops/s	Final attempt (#28)	1,165.0 ops/s, score -0.4568
Bytes read, best (lowest)	270.2 MB (-15.3%)	Refaults, best (lowest)	39,626 (-22.4%)

σ = baseline standard deviation of the 8 noop calibration runs. Δ% is relative to the baseline mean of the same probe. MB = 10^6 bytes.

Summary by attempt class

Class	n	Mean tput	sd	Min	Max	Mean score	Best score	Mean bytes read MB	Mean refaults
real	20	1,234.6	51.4	1,075.1	1,277.7	0.595	0.9980	309.2	48,826
tune	8	1,214.2	71.7	1,070.5	1,273.1	0.393	0.9983	303.9	47,584
all	**28**	**1,228.8**	**57.2**	**1,070.5**	**1,277.7**	**0.537**	**0.9983**	**307.6**	**48,471**

Environment and workload

Evaluation node (CloudLab, Wisconsin)

Component	Value	Component	Value
Node type	c220g1	Kernel	6.6.8-cache-ext+ (Ubuntu 22.04.2)
CPU	2 × Intel Xeon E5-2630 v3 @ 2.40 GHz	Cores / threads	16 cores / 32 threads (2 sockets, 2 NUMA nodes)
CPU max / min MHz; governor	3200 / 1200; schedutil	Caches (L1d / L2 / L3)	512 KiB / 4 MiB / 40 MiB (aggregate)
Memory	131,732,952 kB (125.6 GiB), no swap	Disks	2 × 1.1 TB SAS HDD (ST1200MM0007), 1 × 447 GB SSD (Intel SSDSC2BB480G4)
Benchmark filesystem	/mydata on /dev/sda3 (HDD), 63 GB, 25% used	Cgroup	cgroup v2; benchmark memory.max = 64 MB
Compiler / tools	clang 14.0.0-1ubuntu1.1, bpftool v7.3.0, libbpf (static)	Eval isolation	one eval at a time (flock); policy attached via BPF struct_ops to the cgroup

Workload parameters (benchmark harness defaults, used for every attempt)

Parameter	Value	Parameter	Value
Benchmark	My-YCSB (leveldb-scan branch), run_leveldb	Request distribution / Zipfian constant	zipfian / 0.99
Operation mix	95% GET (read), 5% SCAN	Worker threads	4 (exactly 1 dedicated scan thread)
Scan length	500 keys per SCAN	Measured phase	30 s
Database entries	500,000	Warm-up phase	10 s (same mix, probes not recording)
Key / value size	16 B / 200 B	LevelDB block cache	16 MiB (16,777,216 B)
Cgroup memory limit (page cache)	64 MB	Page cache at start	dropped before each run (echo 3 > drop_caches)
Per-eval timeout	180 s	Observed eval wall time	39.8 s mean (39.6 ... 41.7)
Policy loader	started detached; given -w -s -c	scan_pids map	pinned by harness; policies may read it (ground-truth scan-thread tids)

Scoring and probes

Probe	Source	Direction	Weight	Baseline mean	Baseline sd	CV %	n
throughput (ops/s)	results.json: throughput_ops_per_sec	maximize	2.0	1,224.32	9.52	0.78	8
bytes read (cgroup rbytes)	cgroup io.stat (delta over measured phase)	minimize	0.5	318,859,000	3,868,000	1.21	8
refaults (cgroup memstat)	cgroup memory.stat refault_file (delta)	minimize	0.25	51,054.2	918.7	1.80	8

score = Σ w·tanh(±z) / Σ w, where z = (value − baseline mean)/baseline sd, sign chosen so that an improvement is positive; range [−1, 1], 0 = same as noop. Weights sum to 2.75.

Per-attempt results

#	Local	Commit	Cls	Parent	Score	Tput ops/s	Δ%	z	Bytes read MB	Δ%	Refaults	Δ%	Wall s	Eval s	+/− lines	Code lines
1	00:44:53	3220c3bb	real	49cda6e5	-0.4545	1,075.1	-12.2	-15.7	270.2	-15.3	39,626	-22.4	39.6	373	+509/−50	603
2	00:53:29	7c3ef53d	real	3220c3bb	0.9343	1,263.0	+3.2	+4.1	314.9	-1.2	50,172	-1.7	39.7	46	+409/−410	602
3	00:54:41	995ee7be	real	7c3ef53d	0.8388	1,257.6	+2.7	+3.5	317.0	-0.6	50,720	-0.7	39.6	46	+872/−365	1109
4	00:55:44	1ab2df0b	real	995ee7be	0.2863	1,232.7	+0.7	+0.9	323.3	+1.4	52,177	+2.2	39.6	46	+397/−868	638
5	00:57:13	0e4bf02e	real	1ab2df0b	0.7587	1,237.2	+1.0	+1.4	316.9	-0.6	50,657	-0.8	40.0	46	+13/−38	613
6	00:58:26	9fd454e5	real	0e4bf02e	0.9906	1,261.5	+3.0	+3.9	310.8	-2.5	49,226	-3.6	39.8	46	+1/−11	603
7	01:00:09	b96b2ce0	real	9fd454e5	0.9946	1,256.2	+2.6	+3.4	309.1	-3.1	48,787	-4.4	39.7	46	+1/−2	602
8	01:01:45	4dfced51	real	b96b2ce0	-0.2228	1,216.4	-0.6	-0.8	305.0	-4.3	47,831	-6.3	39.9	46	+1/−1	602
9	01:03:30	a2f787dd	real	4dfced51	0.9852	1,276.4	+4.3	+5.5	311.8	-2.2	49,470	-3.1	39.6	46	+27/−1	628
10	01:05:24	b631bf1b	real	a2f787dd	0.9894	1,273.7	+4.0	+5.2	311.1	-2.4	49,301	-3.4	39.6	46	+1/−1	628
11	01:08:29	2d5d649a	real	b631bf1b	0.8321	1,277.7	+4.4	+5.6	317.2	-0.5	50,729	-0.6	39.9	45	+1/−1	628
12	01:10:29	777cbda4	real	2d5d649a	0.8980	1,237.8	+1.1	+1.4	312.6	-2.0	49,628	-2.8	39.8	46	+9/−4	633
13	01:12:23	c14b63da	real	b631bf1b	-0.4545	1,141.8	-6.7	-8.7	283.0	-11.2	42,637	-16.5	39.8	46	+10/−2	636
14	01:13:50	c01ca9a6	real	c14b63da	0.0326	1,228.2	+0.3	+0.4	324.7	+1.8	52,518	+2.9	39.7	46	+5/−5	636
15	01:15:22	a50fe634	real	c01ca9a6	0.9921	1,255.2	+2.5	+3.2	310.0	-2.8	49,001	-4.0	39.6	46	+0/−3	633
16	01:18:58	5e23d6c7	real	b631bf1b	0.9781	1,247.4	+1.9	+2.4	311.2	-2.4	49,286	-3.5	39.8	46	+6/−9	625
17	01:20:16	61054f8b	real	b631bf1b	0.9980	1,257.4	+2.7	+3.5	305.6	-4.2	48,005	-6.0	39.7	46	+1/−0	629
18	01:23:10	ce24b92d	real	b631bf1b	0.9963	1,258.8	+2.8	+3.6	308.4	-3.3	48,673	-4.7	39.6	46	+17/−2	643
19	01:24:23	ef625089	tune	ce24b92d	-0.4545	1,141.3	-6.8	-8.7	283.7	-11.0	42,743	-16.3	39.8	46	+273/−452	464
20	01:25:39	80c98286	tune	b631bf1b	0.9930	1,273.1	+4.0	+5.1	310.3	-2.7	49,109	-3.8	39.6	46	+1/−0	629
21	01:28:06	23aef7a1	tune	b631bf1b	-0.9040	1,211.4	-1.1	-1.4	327.8	+2.8	53,273	+4.3	39.6	46	+1/−1	628
22	01:30:50	b7395753	real	b631bf1b	0.9832	1,272.9	+4.0	+5.1	312.0	-2.1	49,530	-3.0	39.7	46	+13/−6	635
23	01:33:28	ff01f9d1	tune	b7395753	0.9954	1,254.3	+2.4	+3.1	307.8	-3.5	48,503	-5.0	39.7	46	+3/−2	636
24	01:35:51	60071e54	tune	b631bf1b	0.9769	1,244.7	+1.7	+2.1	308.8	-3.2	48,773	-4.5	39.6	46	+2/−1	629
25	01:37:17	df506481	tune	60071e54	-0.4545	1,070.5	-12.6	-16.1	278.9	-12.5	41,698	-18.3	39.6	46	+476/−319	786
26	01:39:42	e573de75	tune	61054f8b	0.9967	1,259.7	+2.9	+3.7	308.4	-3.3	48,604	-4.8	39.8	46	+3/−3	629
27	01:41:39	50f29a63	tune	e573de75	0.9983	1,258.4	+2.8	+3.6	305.4	-4.2	47,966	-6.0	39.7	46	+1/−0	630
28	01:42:42	c3bd68ae	real	50f29a63	-0.4568	1,165.0	-4.8	-6.2	308.2	-3.4	48,549	-4.9	41.7	48	+1/−0	631

Local = submit time. Eval s = submit → graded (attempt #1 includes the one-time 8-run noop calibration). +/− lines = diff of noop.c against the parent attempt. z = throughput z-score vs baseline.

Score composition per attempt

#	Commit	z tput	contrib tput (w=2.0)	z bytes	contrib bytes (w=0.5)	z refault	contrib refault (w=0.25)	weighted sum	score
1	3220c3bb	-15.67	-1.000	-12.57	+1.000	-12.44	+1.000	-1.250	-0.4545
2	7c3ef53d	+4.06	+0.999	-1.02	+0.769	-0.96	+0.744	+2.568	0.9343
3	995ee7be	+3.50	+0.998	-0.48	+0.446	-0.36	+0.349	+2.306	0.8388
4	1ab2df0b	+0.88	+0.704	+1.16	-0.821	+1.22	-0.840	+0.787	0.2863
5	0e4bf02e	+1.35	+0.874	-0.52	+0.474	-0.43	+0.407	+2.087	0.7587
6	9fd454e5	+3.91	+0.999	-2.09	+0.970	-1.99	+0.963	+2.724	0.9906
7	b96b2ce0	+3.35	+0.998	-2.53	+0.987	-2.47	+0.986	+2.736	0.9946
8	4dfced51	-0.83	-0.681	-3.58	+0.998	-3.51	+0.998	-0.614	-0.2228
9	a2f787dd	+5.47	+1.000	-1.83	+0.950	-1.72	+0.938	+2.710	0.9852
10	b631bf1b	+5.19	+1.000	-1.99	+0.964	-1.91	+0.957	+2.721	0.9894
11	2d5d649a	+5.60	+1.000	-0.43	+0.407	-0.35	+0.340	+2.288	0.8321
12	777cbda4	+1.42	+0.889	-1.62	+0.925	-1.55	+0.914	+2.469	0.8980
13	c14b63da	-8.67	-1.000	-9.26	+1.000	-9.16	+1.000	-1.250	-0.4545
14	c01ca9a6	+0.41	+0.387	+1.51	-0.907	+1.59	-0.921	+0.090	0.0326
15	a50fe634	+3.25	+0.997	-2.29	+0.980	-2.24	+0.977	+2.728	0.9921
16	5e23d6c7	+2.42	+0.984	-1.98	+0.963	-1.92	+0.958	+2.689	0.9781
17	61054f8b	+3.48	+0.998	-3.43	+0.998	-3.32	+0.997	+2.744	0.9980
18	ce24b92d	+3.62	+0.999	-2.69	+0.991	-2.59	+0.989	+2.741	0.9963
19	ef625089	-8.72	-1.000	-9.09	+1.000	-9.05	+1.000	-1.250	-0.4545
20	80c98286	+5.12	+1.000	-2.20	+0.976	-2.12	+0.971	+2.731	0.9930
21	23aef7a1	-1.35	-0.875	+2.31	-0.980	+2.42	-0.984	-2.486	-0.9040
22	b7395753	+5.10	+1.000	-1.76	+0.943	-1.66	+0.930	+2.704	0.9832
23	ff01f9d1	+3.15	+0.996	-2.85	+0.993	-2.78	+0.992	+2.737	0.9954
24	60071e54	+2.14	+0.973	-2.61	+0.989	-2.48	+0.986	+2.687	0.9769
25	df506481	-16.15	-1.000	-10.34	+1.000	-10.18	+1.000	-1.250	-0.4545
26	e573de75	+3.72	+0.999	-2.72	+0.991	-2.67	+0.990	+2.741	0.9967
27	50f29a63	+3.58	+0.998	-3.48	+0.998	-3.36	+0.998	+2.744	0.9983
28	c3bd68ae	-6.24	-1.000	-2.76	+0.992	-2.73	+0.991	-1.256	-0.4568

weighted sum / 2.75 = score. Contributions are tanh-squashed, so |z| ≳ 2 gives ≈ ±0.96 or more.

Derived per-operation metrics

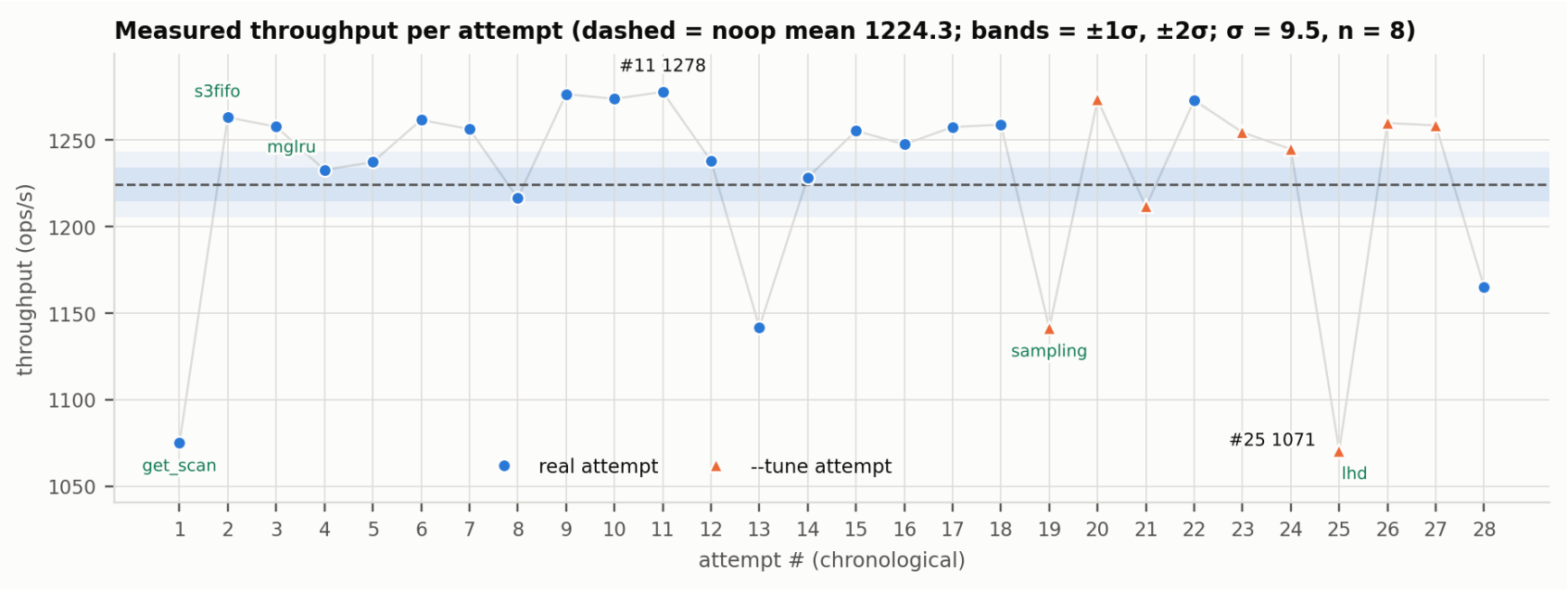
#	Commit	Cls	Ops in measured phase (≈ tput × 30 s)	Bytes read per op (KB)	Refaults per 1k ops	Read rate (MB/s)	Refaults per MB read	Wall s
1	3220c3bb	real	32,254	8.38	1,228.56	9.01	146.64	39.6
2	7c3ef53d	real	37,889	8.31	1,324.18	10.50	159.31	39.7
3	995ee7be	real	37,728	8.40	1,344.35	10.57	160.00	39.6
4	1ab2df0b	real	36,980	8.74	1,410.97	10.78	161.37	39.6
5	0e4bf02e	real	37,115	8.54	1,364.87	10.56	159.87	40.0
6	9fd454e5	real	37,846	8.21	1,300.71	10.36	158.40	39.8
7	b96b2ce0	real	37,686	8.20	1,294.57	10.30	157.84	39.7
8	4dfced51	real	36,492	8.36	1,310.71	10.17	156.83	39.9
9	a2f787dd	real	38,292	8.14	1,291.92	10.39	158.67	39.6
10	b631bf1b	real	38,211	8.14	1,290.23	10.37	158.45	39.6
11	2d5d649a	real	38,330	8.28	1,323.49	10.57	159.93	39.9
12	777cbda4	real	37,135	8.42	1,336.44	10.42	158.77	39.8
13	c14b63da	real	34,253	8.26	1,244.75	9.43	150.65	39.8
14	c01ca9a6	real	36,846	8.81	1,425.34	10.82	161.75	39.7
15	a50fe634	real	37,657	8.23	1,301.24	10.33	158.07	39.6
16	5e23d6c7	real	37,422	8.32	1,317.04	10.37	158.38	39.8
17	61054f8b	real	37,723	8.10	1,272.57	10.19	157.10	39.7
18	ce24b92d	real	37,762	8.17	1,288.92	10.28	157.80	39.6
19	ef625089	tune	34,238	8.29	1,248.42	9.46	150.67	39.8
20	80c98286	tune	38,193	8.13	1,285.80	10.34	158.25	39.6
21	23aef7a1	tune	36,343	9.02	1,465.84	10.93	162.52	39.6
22	b7395753	real	38,187	8.17	1,297.05	10.40	158.73	39.7
23	ff01f9d1	tune	37,628	8.18	1,289.01	10.26	157.56	39.7
24	60071e54	tune	37,341	8.27	1,306.16	10.29	157.95	39.6
25	df506481	tune	32,116	8.68	1,298.37	9.30	149.53	39.6
26	e573de75	tune	37,791	8.16	1,286.12	10.28	157.63	39.8
27	50f29a63	tune	37,751	8.09	1,270.60	10.18	157.07	39.7
28	c3bd68ae	real	34,948	8.82	1,389.16	10.27	157.54	41.7
noop	(baseline)	-	36,730	8.68	1,390.00	10.63	160.12	-

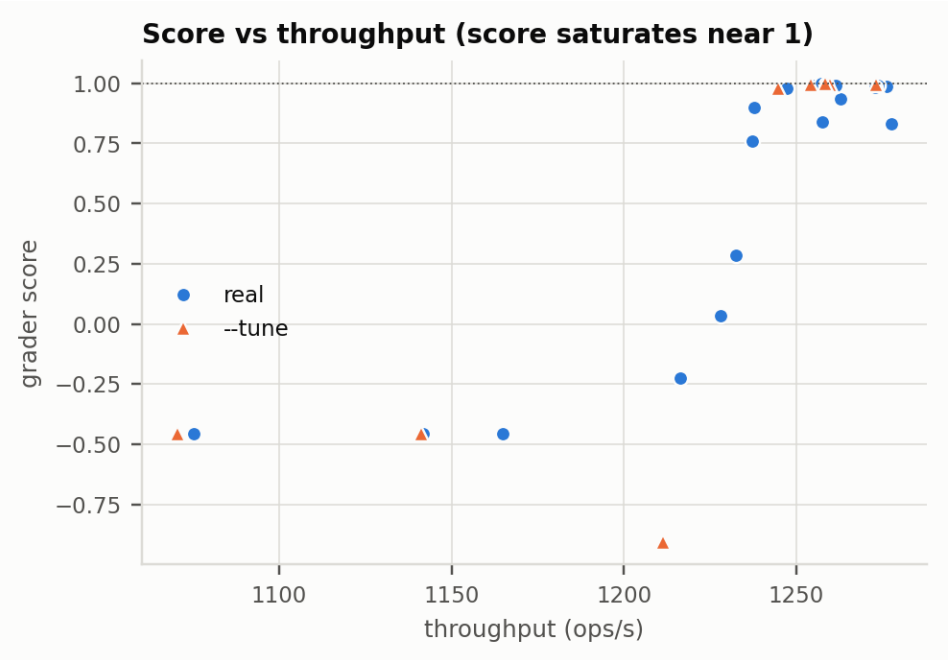
Assumes the probes cover exactly the 30 s measured phase (the harness wraps probes around that phase only), so operations = throughput × 30 and bytes/refaults are per measured phase. Refault = a page evicted earlier and read back in.

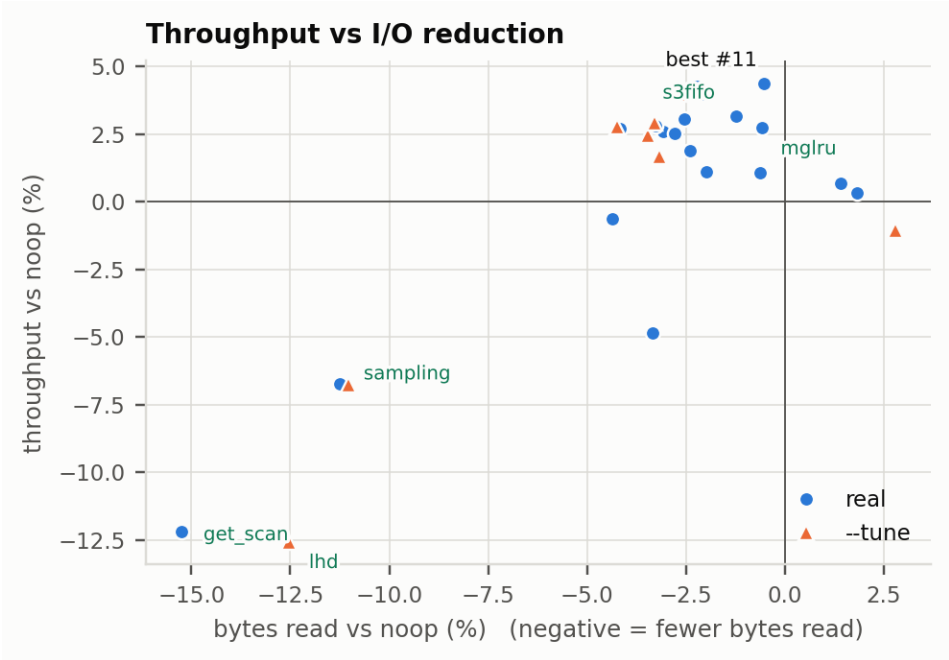
Attempt descriptions (commit messages, verbatim)

#	Commit	Cls	Description
1	3220c3bb	real	baseline: reference get_scan.c unchanged (scan-pid list + sampled LFU w/ index-block bonus) to calibrate where references land
2	7c3ef53d	real	reference s3fifo.c unchanged (landscape mapping)
3	995ee7be	real	reference mglru.c unchanged (landscape mapping)
4	1ab2df0b	real	s3fifo + scan_pids: scan-thread accesses don't bump freq (sequential scans touch each page many times), scan-added pages bypass ghost check and get no ghost entry unless a GET touched them
5	0e4bf02e	real	s3fifo + correlated-reference period: accesses within 2ms of the last counted one don't bump freq (filters in-scan repeat touches of the same page, keeps cross-scan/GET reuse)
6	9fd454e5	real	noise check: plain s3fifo re-run (identical to 7c3ef53d except trailing comment)
7	b96b2ce0	real	s3fifo with small queue 1/10 of cache instead of 1/15 (canonical S3-FIFO ratio)
8	4dfced51	real	s3fifo small 1/10 + ghost map 2x cache size (remember more evicted pages so quick re-references go straight to main)
9	a2f787dd	real	s3fifo 1/10 + ghost 2x, but scan-thread re-adds never promote via ghost (test: does scan ghost-flooding of main cause the eval-8 throughput loss?)
10	b631bf1b	real	gated-ghost s3fifo with ghost 4x cache (only GET-thread re-adds use ghost, so remember more GET reuse)
11	2d5d649a	real	gated-ghost s3fifo (ghost 4x) with small queue 1/8 of cache (bigger small queue cut I/O in eval 7; throughput has slack now)
12	777cbda4	real	gated-ghost s3fifo (4x, small 1/10): scan-thread ghost hits now enter main at the HEAD (low priority) instead of staying in small; GET ghost hits still main tail
13	c14b63da	real	structural attempt 1/3 on GET-aware aging: gated-ghost s3fifo (4x, 1/10) + get_touched bit; main CLOCK decrements scan-only pages by 2, GET-touched by 1
14	c01ca9a6	real	structural attempt 2/3 on GET-aware aging: milder penalty - scan-only pages freq cap 2 (GET-touched cap 3), normal decrement
15	a50fe634	real	structural attempt 3/3 on GET-aware aging: cap-2 for scan-only pages, but get_touched set ONLY at add time (no scan_pids lookup in folio_accessed) - tests hook-overhead hypothesis
16	5e23d6c7	real	hook-overhead attempt 1/3: gated-ghost s3fifo (b631bf1b) with cheaper folio_accessed - drop watchlist lookup (metadata lookup implies relevance), plain read/store instead of atomic, no-op at cap 3
17	61054f8b	real	drift check: re-run gated-ghost s3fifo b631bf1b unchanged (eval16's cheaper hook gave 1247 vs 1274 ops/s; is that real or machine drift?)
18	ce24b92d	real	gated-ghost s3fifo (b631bf1b) + LevelDB index-block protection: last page of each file promoted from small to main and skipped by main CLOCK eviction (eviction-path only, hot hook unchanged)
19	ef625089	tune	tune probe: reference sampling.c unchanged (landscape point + learn what --tune does on this grader)
20	80c98286	tune	variance check: 3rd run of gated-ghost s3fifo b631bf1b (runs so far: 1273.7/311.1MB, 1257.4/305.6MB)
21	23aef7a1	tune	ablation: gated-ghost s3fifo (small 1/10) with ghost back to 1x cache - is the bigger ghost load-bearing or is the thread gate alone enough?
22	b7395753	real	gated-ghost s3fifo, ghost 2x, scan-added pages that never reached main get NO ghost entry (eval21 ablation: 1x gated ghost = -0.90, suggesting scan evictions crowd GET history out of the ghost)
23	ff01f9d1	tune	b7395753 + GET-added pages promote small->main on first reuse (freq>0); scan-added pages still need freq>1
24	60071e54	tune	gated-ghost s3fifo b631bf1b with ghost map as common (global) LRU instead of BPF_F_NO_COMMON_LRU: per-CPU partitions may leave most ghost capacity unused when kswapd does the evicting
25	df506481	tune	landscape: reference lhd.c unchanged (last unmapped reference policy)
26	e573de75	tune	gated-ghost s3fifo (61054f8b) with freq cap 2 instead of 3: fewer CLOCK passes in main, faster turnover of formerly-hot pages
27	50f29a63	tune	repeat #2 of freq-cap-2 gated-ghost s3fifo (e573de75: 1259.7 ops/s, 308.4 MB) to compare mean vs b631bf1b (3 runs: mean 1268 ops/s, 309.0 MB)
28	c3bd68ae	real	real submission of freq-cap-2 gated-ghost s3fifo (2 tune runs: 0.9967 / 0.9983, mean 306.9 MB vs base 309.0 MB; throughput saturated at ~1259)

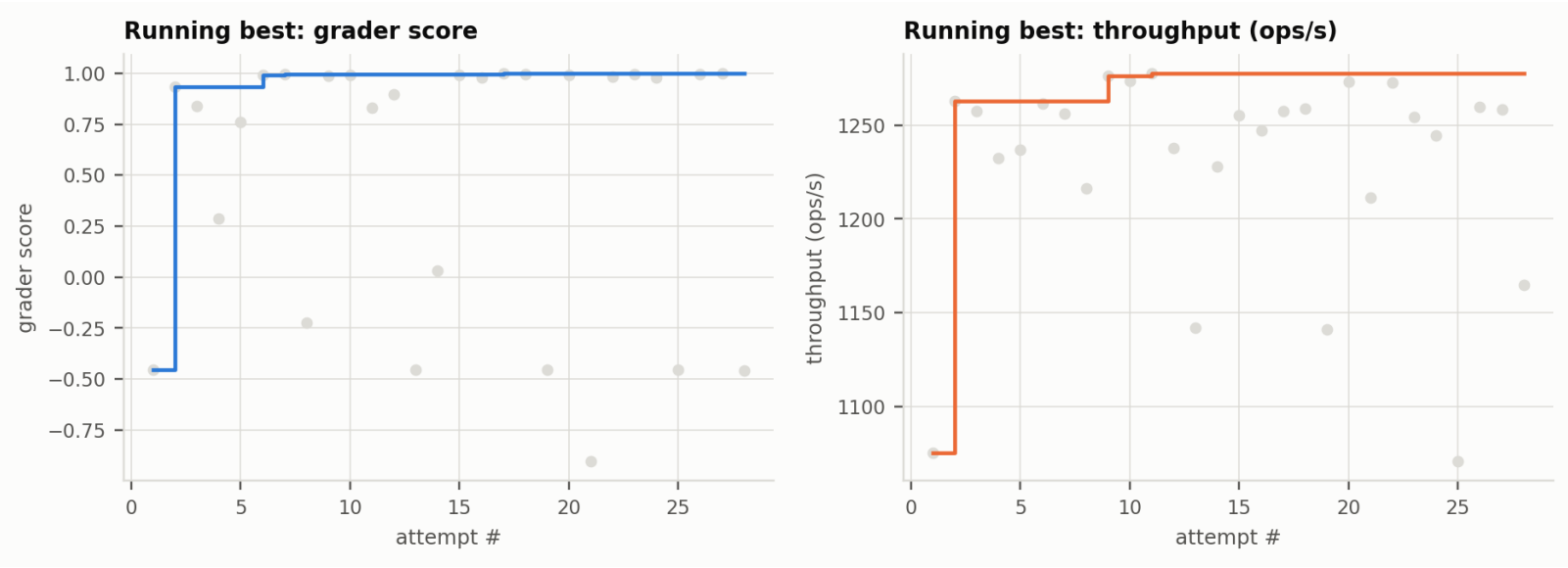
Throughput over the run

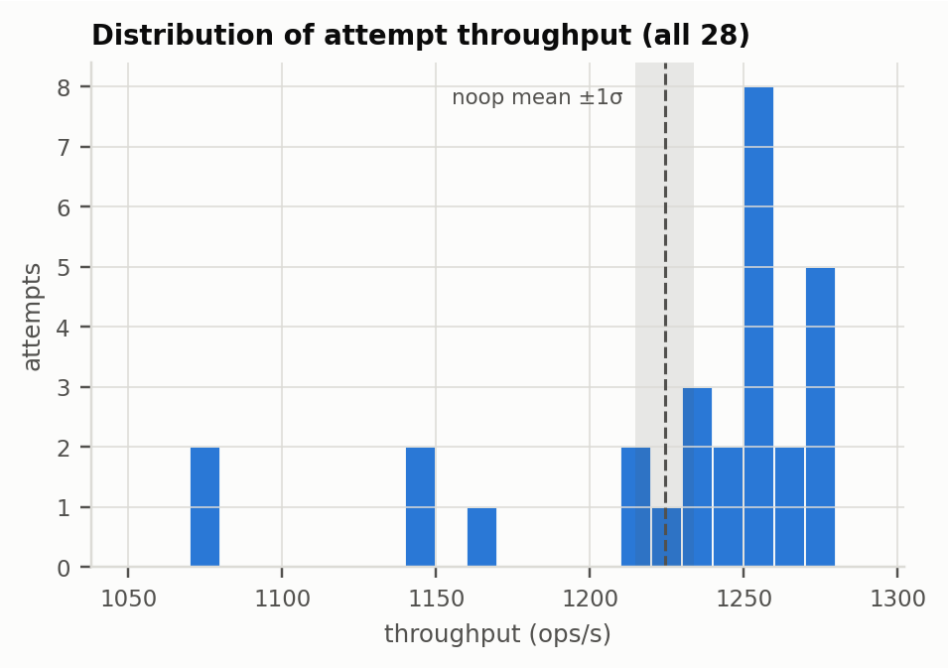


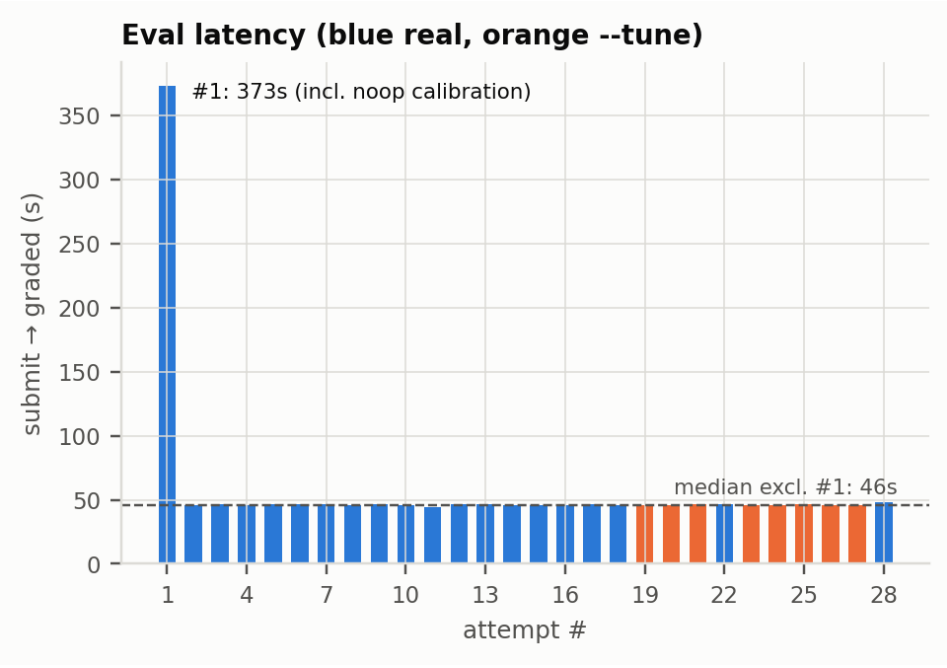




Progress and distribution







Reference policies run unmodified

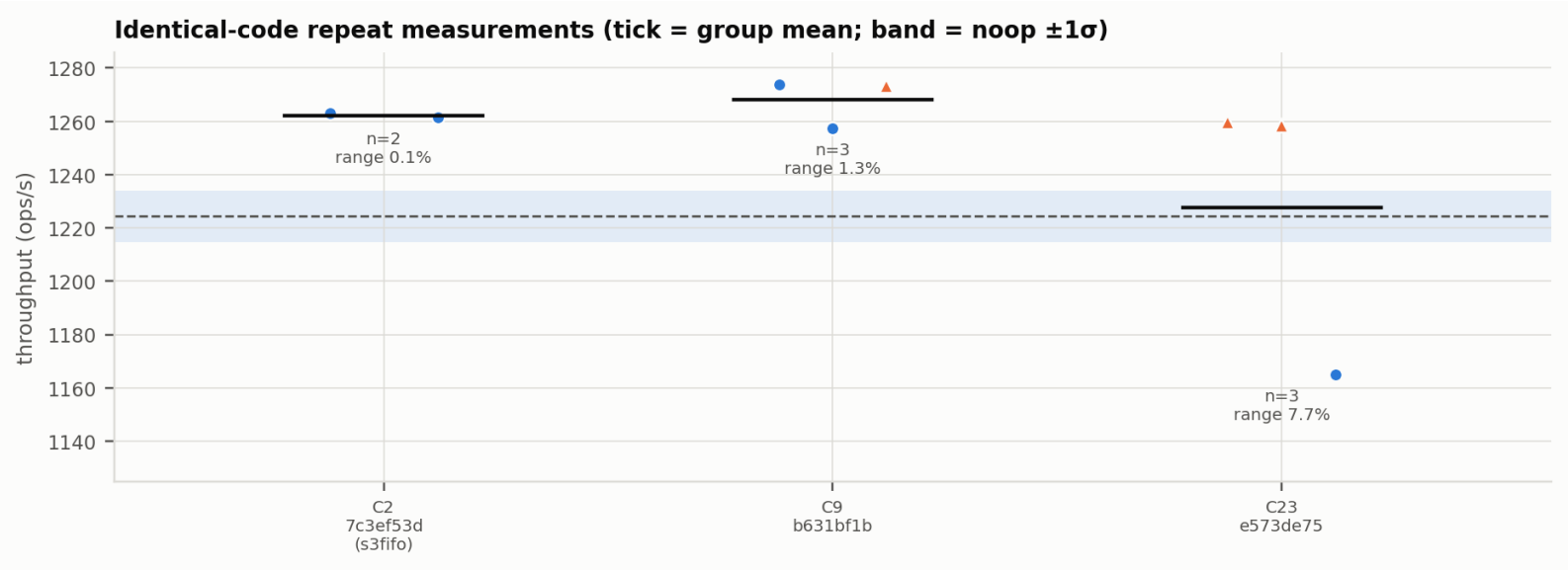
#	Reference policy	Cls	Score	Tput ops/s	Δ% vs noop	Bytes read MB	Δ%	Refaults	Δ%
1	get_scan.c	real	-0.4545	1,075.1	-12.19	270.2	-15.3	39,626	-22.4
2	s3fifo.c	real	0.9343	1,263.0	+3.16	314.9	-1.2	50,172	-1.7
3	mglru.c	real	0.8388	1,257.6	+2.72	317.0	-0.6	50,720	-0.7
19	sampling.c	tune	-0.4545	1,141.3	-6.78	283.7	-11.0	42,743	-16.3
25	lhd.c	tune	-0.4545	1,070.5	-12.56	278.9	-12.5	41,698	-18.3

fifo.c and mru.c were not evaluated in this run (they were only smoke-tested before it).

Repeat measurements of identical code

Group	First #	Attempts (short hash, class r/t)	n	Mean	sd	Min	Max	Range %	CV %	Mean Δ% vs noop	Mean bytes MB	Mean refaults	Mean score
C2	2	#2 7c3ef53dr, #6 9fd454e5r	2	1,262.2	1.0	1,261.5	1,263.0	0.11	0.08	+3.10	312.8	49,699	0.962
C9	10	#10 b631bf1br, #17 61054f8br, #20 80c98286t	3	1,268.1	9.2	1,257.4	1,273.7	1.28	0.73	+3.57	309.0	48,805	0.993
C23	26	#26 e573de75t, #27 50f29a63t, #28 c3bd68aer	3	1,227.7	54.3	1,165.0	1,259.7	7.72	4.42	+0.27	307.3	48,373	0.513

Noise estimate	Value
Groups with ≥ 2 identical-code measurements	3 groups, 8 measurements
Pooled within-group sd (throughput)	34.85 ops/s (df = 5; CV 2.85%)
Baseline sd (8 consecutive noop runs)	9.52 ops/s (CV 0.78%)
Ratio pooled / baseline sd	3.66
Pooled sd excluding the most deviant single measurement (#28 c3bd68ae, 1,165.0 ops/s)	6.56 ops/s (df = 4; CV 0.54%; ratio to baseline sd 0.69)
Largest within-group range	7.72 % of group mean
Smallest 95% detectable difference between two means of 3 evals each (pooled sd, all data / excluding that outlier)	≈ 73.2 ops/s (5.98%) / ≈ 14.9 ops/s (1.21%)

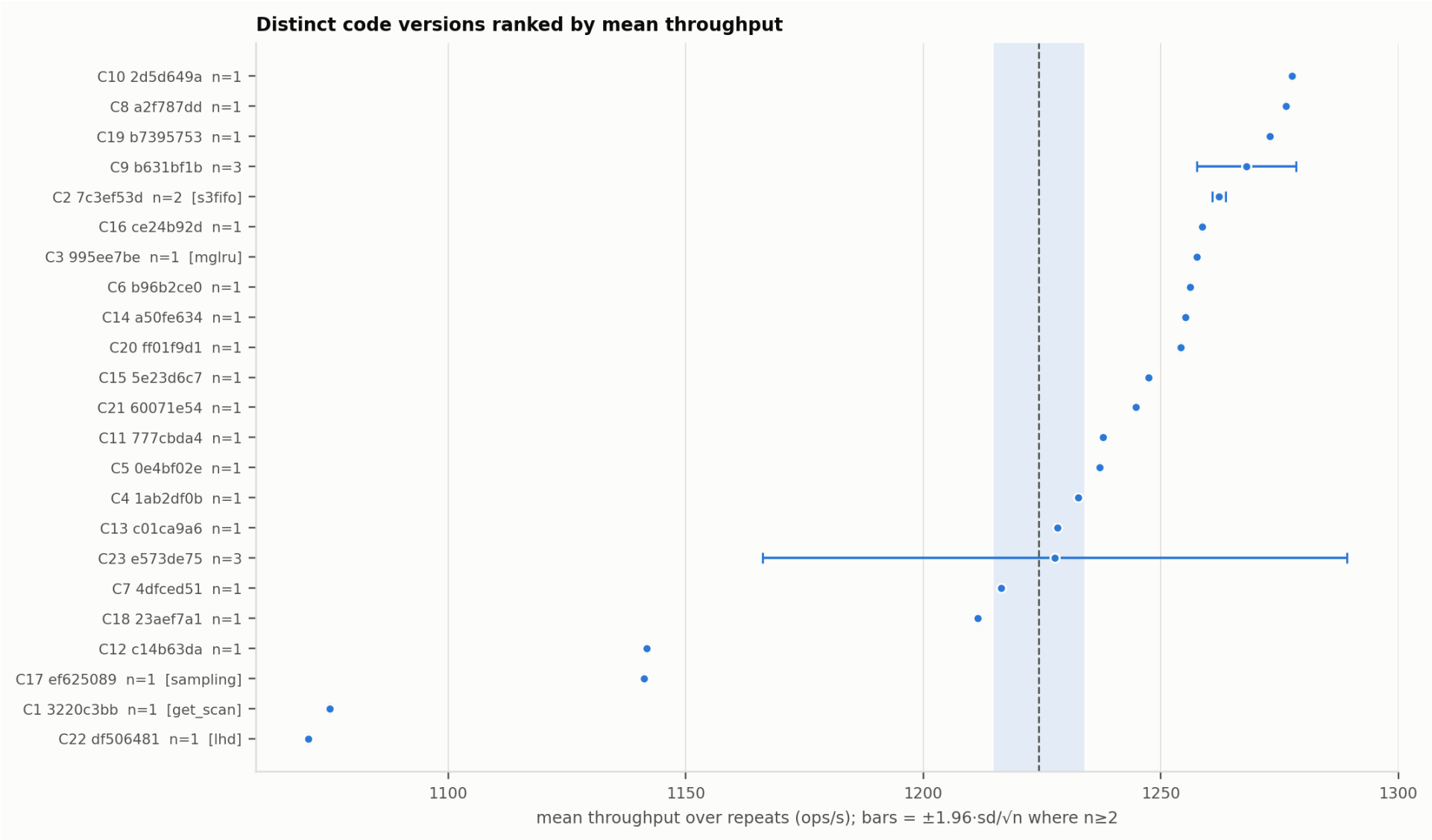


Distinct code versions: ranking by mean throughput

Rank	Group	First #	n	Mean tput	sd	Δ% vs noop	Best single tput	Mean bytes MB	Mean refaults	Mean score	Best single score	Ref	Description (first attempt)
1	C10	11	1	1,277.7	-	+4.36	1,277.7	317.2	50,729	0.832	0.832		gated-ghost s3fifo (ghost 4x) with small queue 1/8 of cache (bigger small queue cut I/O in eval 7; throughput has slack now)
2	C8	9	1	1,276.4	-	+4.25	1,276.4	311.8	49,470	0.985	0.985		s3fifo 1/10 + ghost 2x, but scan-thread re-adds never promote via ghost (test: does scan ghost-flooding of main cause the eval-8 throughput loss?)
3	C19	22	1	1,272.9	-	+3.97	1,272.9	312.0	49,530	0.983	0.983		gated-ghost s3fifo, ghost 2x, scan-added pages that never reached main get NO ghost entry (eval21 ablation: 1x gated ghost = -0.90, suggesting scan evictions crowd GET history out of the ghost)
4	C9	10	3	1,268.1	9.2	+3.57	1,273.7	309.0	48,805	0.993	0.998		gated-ghost s3fifo with ghost 4x cache (only GET-thread re-adds use ghost, so remember more GET reuse)
5	C2	2	2	1,262.2	1.0	+3.10	1,263.0	312.8	49,699	0.962	0.991	s3fifo	reference s3fifo.c unchanged (landscape mapping)
6	C16	18	1	1,258.8	-	+2.81	1,258.8	308.4	48,673	0.996	0.996		gated-ghost s3fifo (b631bf1b) + LevelDB index-block protection: last page of each file promoted from small to main and skipped by main CLOCK eviction (eviction-path only, hot hook unchanged)
7	C3	3	1	1,257.6	-	+2.72	1,257.6	317.0	50,720	0.839	0.839	mglru	reference mglru.c unchanged (landscape mapping)
8	C6	7	1	1,256.2	-	+2.60	1,256.2	309.1	48,787	0.995	0.995		s3fifo with small queue 1/10 of cache instead of 1/15 (canonical S3-FIFO ratio)
9	C14	15	1	1,255.2	-	+2.53	1,255.2	310.0	49,001	0.992	0.992		structural attempt 3/3 on GET-aware aging: cap-2 for scan-only pages, but get_touched set ONLY at add time (no scan_pids lookup in folio_accessed) - tests hook-overhead hypothesis
10	C20	23	1	1,254.3	-	+2.45	1,254.3	307.8	48,503	0.995	0.995		b7395753 + GET-added pages promote small->main on first reuse (freq>0); scan-added pages still need freq>1
11	C15	16	1	1,247.4	-	+1.88	1,247.4	311.2	49,286	0.978	0.978		hook-overhead attempt 1/3: gated-ghost s3fifo (b631bf1b) with cheaper folio_accessed - drop watchlist lookup (metadata lookup implies relevance), plain read/store instead of atomic, no-op at cap 3
12	C21	24	1	1,244.7	-	+1.66	1,244.7	308.8	48,773	0.977	0.977		gated-ghost s3fifo b631bf1b with ghost map as common (global) LRU instead of BPF_F_NO_COMMON_LRU: per-CPU partitions may leave most ghost capacity unused when kswapd does the evicting
13	C11	12	1	1,237.8	-	+1.10	1,237.8	312.6	49,628	0.898	0.898		gated-ghost s3fifo (4x, small 1/10): scan-thread ghost hits now enter main at the HEAD (low priority) instead of staying in small; GET ghost hits still main tail
14	C5	5	1	1,237.2	-	+1.05	1,237.2	316.9	50,657	0.759	0.759		s3fifo + correlated-reference period: accesses within 2ms of the last counted one don't bump freq (filters in-scan repeat touches of the same page, keeps cross-scan/GET reuse)
15	C4	4	1	1,232.7	-	+0.68	1,232.7	323.3	52,177	0.286	0.286		s3fifo + scan_pids: scan-thread accesses don't bump freq (sequential scans touch each page many times), scan-added pages bypass ghost check and get no ghost entry unless a GET touched them
16	C13	14	1	1,228.2	-	+0.32	1,228.2	324.7	52,518	0.033	0.033		structural attempt 2/3 on GET-aware aging: milder penalty - scan-only pages freq cap 2 (GET-touched cap 3), normal decrement
17	C23	26	3	1,227.7	54.3	+0.27	1,259.7	307.3	48,373	0.513	0.998		gated-ghost s3fifo (61054f8b) with freq cap 2 instead of 3: fewer CLOCK passes in main, faster turnover of formerly-hot pages
18	C7	8	1	1,216.4	-	-0.65	1,216.4	305.0	47,831	-0.223	-0.223		s3fifo small 1/10 + ghost map 2x cache size (remember more evicted pages so quick re-references go straight to main)
19	C18	21	1	1,211.4	-	-1.05	1,211.4	327.8	53,273	-0.904	-0.904		ablation: gated-ghost s3fifo (small 1/10) with ghost back to 1x cache - is the bigger ghost load-bearing or is the thread gate alone enough?
20	C12	13	1	1,141.8	-	-6.74	1,141.8	283.0	42,637	-0.455	-0.455		structural attempt 1/3 on GET-aware aging: gated-ghost s3fifo (4x, 1/10) + get_touched bit; main CLOCK decrements scan-only pages by 2, GET-touched by 1
21	C17	19	1	1,141.3	-	-6.78	1,141.3	283.7	42,743	-0.455	-0.455	sampling	tune probe: reference sampling.c unchanged (landscape point + learn what --tune does on this grader)

Rank	Group	First #	n	Mean tput	sd	Δ% vs noop	Best single tput	Mean bytes MB	Mean refaults	Mean score	Best single score	Ref	Description (first attempt)
22	C1	1	1	1,075.1	-	-12.19	1,075.1	270.2	39,626	-0.455	-0.455	get_scan	baseline: reference get_scan.c unchanged (scan-pid list + sampled LFU w/ index-block bonus) to calibrate where references land
23	C22	25	1	1,070.5	-	-12.56	1,070.5	278.9	41,698	-0.455	-0.455	lhd	landscape: reference lhd.c unchanged (last unmapped reference policy)

Ranking by mean throughput (graphical)



Each code version vs the unmodified reference s3fifo.c (pooled-sd Student t, 95% CI)

Group	First #	n	Mean tput	Δ vs ref (ops/s)	Δ vs ref (%)	SE (pooled sd, all)	95% CI low	95% CI high	excl. 0	SE (outlier dropped)	95% CI low	95% CI high	excl. 0
C10	11	1	1,277.7	+15.4	+1.22	42.7	-94.3	+125.2	no	8.0	-6.9	+37.7	no
C8	9	1	1,276.4	+14.1	+1.12	42.7	-95.6	+123.9	no	8.0	-8.2	+36.5	no
C19	22	1	1,272.9	+10.6	+0.84	42.7	-99.1	+120.4	no	8.0	-11.7	+33.0	no
C9	10	3	1,268.1	+5.8	+0.46	31.8	-76.0	+87.6	no	6.0	-10.8	+22.5	no
C16	18	1	1,258.8	-3.5	-0.28	42.7	-113.2	+106.3	no	8.0	-25.8	+18.8	no
C3	3	1	1,257.6	-4.6	-0.37	42.7	-114.4	+105.1	no	8.0	-26.9	+17.7	no
C6	7	1	1,256.2	-6.0	-0.48	42.7	-115.8	+103.7	no	8.0	-28.4	+16.3	no
C14	15	1	1,255.2	-7.0	-0.55	42.7	-116.8	+102.7	no	8.0	-29.3	+15.3	no
C20	23	1	1,254.3	-8.0	-0.63	42.7	-117.7	+101.8	no	8.0	-30.3	+14.3	no
C15	16	1	1,247.4	-14.9	-1.18	42.7	-124.6	+94.9	no	8.0	-37.2	+7.5	no
C21	24	1	1,244.7	-17.6	-1.39	42.7	-127.3	+92.2	no	8.0	-39.9	+4.8	no
C11	12	1	1,237.8	-24.4	-1.94	42.7	-134.2	+85.3	no	8.0	-46.7	-2.1	yes
C5	5	1	1,237.2	-25.1	-1.99	42.7	-134.8	+84.7	no	8.0	-47.4	-2.8	yes
C4	4	1	1,232.7	-29.6	-2.34	42.7	-139.3	+80.2	no	8.0	-51.9	-7.3	yes
C13	14	1	1,228.2	-34.0	-2.70	42.7	-143.8	+75.7	no	8.0	-56.4	-11.7	yes
C23	26	3	1,227.7	-34.6	-2.74	31.8	-116.4	+47.2	no	6.6	-21.4	+15.0	no
C7	8	1	1,216.4	-45.8	-3.63	42.7	-155.6	+63.9	no	8.0	-68.1	-23.5	yes
C18	21	1	1,211.4	-50.8	-4.03	42.7	-160.6	+58.9	no	8.0	-73.1	-28.5	yes
C12	13	1	1,141.8	-120.5	-9.54	42.7	-230.2	-10.7	yes	8.0	-142.8	-98.2	yes
C17	19	1	1,141.3	-121.0	-9.58	42.7	-230.7	-11.2	yes	8.0	-143.3	-98.7	yes
C1	1	1	1,075.1	-187.1	-14.82	42.7	-296.9	-77.4	yes	8.0	-209.4	-164.8	yes
C22	25	1	1,070.5	-191.7	-15.19	42.7	-301.5	-82.0	yes	8.0	-214.0	-169.4	yes

Reference s3fifo.c: n = 2, mean 1,262.2 ops/s. SE = pooled sd × √(1/n + 1/n_ref). Left: pooled sd 34.85 ops/s (df 5, t = 2.571); right: pooled sd 6.56 ops/s with the most deviant measurement dropped (df 4, t = 2.776). n = 1 groups use the same pooled sd. In the right-hand columns the dropped measurement (#28 c3bd68ae) is also removed from its own group's mean, and the CIs are for the difference of the remaining means (the Δ columns on the left are the unadjusted means).

Score analysis

Correlations across the 28 attempts

Pearson r	score	throughput	bytes read	refaults	wallclock	Spearman ρ	score	throughput	bytes read	refaults	wallclock
score	1.000	0.825	0.416	0.416	-0.290	score	1.000	0.703	-0.073	-0.072	-0.075
throughput	0.825	1.000	0.788	0.788	-0.183	throughput	0.703	1.000	0.367	0.374	-0.057
bytes read	0.416	0.788	1.000	1.000	0.032	bytes read	-0.073	0.367	1.000	0.999	0.003
refaults	0.416	0.788	1.000	1.000	0.029	refaults	-0.072	0.374	0.999	1.000	-0.006
wallclock	-0.290	-0.183	0.032	0.029	1.000	wallclock	-0.075	-0.057	0.003	-0.006	1.000

Score saturation

Metric	Value
Attempts with score ≥ 0.95	14 of 28
Throughput span of those attempts	1,244.7 … 1,276.4 ops/s (31.7 ops/s, 2.59% of baseline)
Score span of those attempts	0.9769 … 0.9983 (0.0214)
Attempts with score ≤ −0.40	6 of 28 (throughput 1,070.5 … 1,211.4 ops/s)
Attempts at the score floor (within 0.005 of −0.4545 = (−2 + 0.5 + 0.25)/2.75: throughput saturated at −1, I/O probes at +1)	5

Timing

Run phases (local time)

Event	Time	Elapsed since start (min)
Run directory created / manager started	00:40:13	0.0
First model request at the gateway	00:40:23	0.2
First submission (attempt #1)	00:44:53	4.7
Attempt #1 graded (includes noop calibration)	00:51:06	10.9
Last submission (attempt #28)	01:42:42	62.5
Last attempt graded / auto-stop recorded	01:43:33	63.3

Evaluation latency (submit → graded)

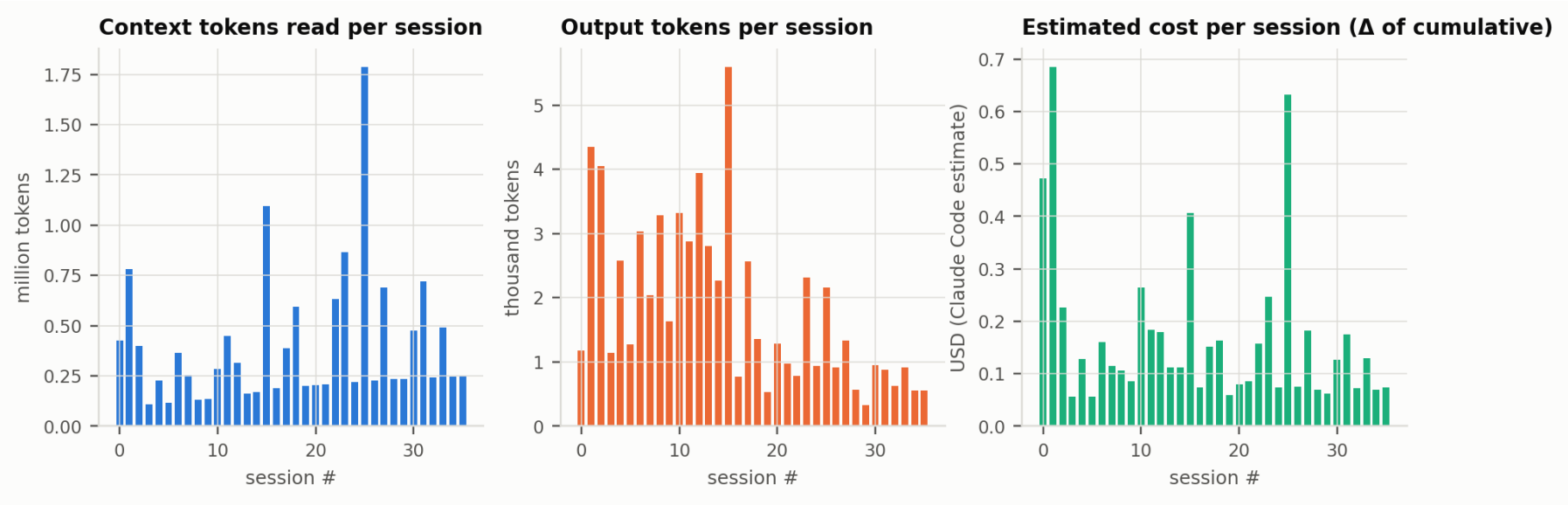
Subset	n	Mean s	Median s	p90 s	Min s	Max s	Total s
all attempts	28	57.8	46.1	46.3	44.5	373.0	1,617
all excluding #1 (calibration)	27	46.1	46.1	46.3	44.5	48.0	1,244
real, excl. #1	19	46.1	46.2	46.3	44.5	48.0	876
tune	8	46.1	46.1	46.2	45.8	46.2	368

Time budget	Seconds	% of run wall time
Evaluation (attempts #2–#28, submit → graded)	1,244	32.7
Attempt #1 incl. calibration	373	9.8
Everything else (agent reading, thinking, editing, note-writing)	2,184	57.5
Run wall time	**3,801**	**100.0**

Agent sessions and LLM usage

#	Start	Wall s	API s (Δ)	Turns	Cost $ (Δ)	Input tok	Cache write tok	Cache read tok	Output tok	Thinking tok	API retries	Denials	Terminal reason
0	00:40:21	645	159	9	0.472	14	75,706	350,843	1,182	200	0	0	aborted_streaming
1	00:51:07	190	127	11	0.686	20	92,182	689,010	4,347	1,563	0	2	aborted_streaming
2	00:54:17	70	53	6	0.226	10	13,492	385,647	4,052	911	0	0	aborted_streaming
3	00:55:27	65	13	3	0.056	4	2,500	105,674	1,137	0	0	0	aborted_streaming
4	00:56:33	90	32	4	0.128	6	6,484	219,468	2,578	691	0	0	aborted_streaming
5	00:58:03	70	19	3	0.057	4	1,695	115,208	1,270	97	0	0	aborted_streaming
6	00:59:13	105	42	5	0.160	8	5,505	358,004	3,027	483	0	0	aborted_streaming
7	01:00:58	95	37	4	0.115	6	4,878	248,103	2,043	64	0	0	aborted_streaming
8	01:02:34	105	52	3	0.106	4	2,889	127,286	3,285	1,247	0	0	aborted_streaming
9	01:04:19	115	36	3	0.085	4	5,256	130,175	1,632	226	0	0	aborted_streaming
10	01:06:14	180	68	5	0.264	6	8,020	276,176	3,317	378	0	0	aborted_streaming
11	01:09:14	125	53	5	0.184	8	7,658	440,874	2,881	386	0	0	aborted_streaming
12	01:11:19	110	54	4	0.179	6	7,841	306,767	3,936	1,448	0	0	aborted_streaming
13	01:13:10	90	36	3	0.112	4	4,899	158,950	2,803	919	0	0	aborted_streaming
14	01:14:40	90	38	3	0.111	4	6,652	163,849	2,267	563	0	0	aborted_streaming
15	01:16:10	215	115	10	0.406	14	15,552	1,080,697	5,598	831	0	3	aborted_streaming
16	01:19:46	80	14	5	0.074	4	4,210	186,053	766	289	0	0	aborted_streaming
17	01:21:06	58	57	2	0.151	6	4,652	383,382	2,564	219	0	1	completed
18	01:22:06	115	29	6	0.163	8	3,671	589,872	1,353	207	0	0	aborted_streaming
19	01:24:01	70	10	3	0.059	2	1,726	199,127	533	218	0	0	aborted_streaming
20	01:25:11	75	22	5	0.079	4	2,692	200,853	1,287	763	0	0	aborted_streaming
21	01:26:27	13	13	1	0.086	4	5,146	203,545	968	164	0	0	completed
22	01:26:42	135	32	7	0.158	8	3,152	630,889	788	100	0	0	aborted_streaming
23	01:28:57	160	54	7	0.247	10	5,768	858,898	2,308	668	0	0	aborted_streaming
24	01:31:37	15	15	1	0.073	4	2,179	217,611	941	159	0	0	completed
25	01:31:57	140	81	10	0.633	18	5,528	1,783,133	2,162	603	0	0	aborted_streaming
26	01:34:17	15	15	1	0.075	4	2,255	225,318	913	55	0	0	completed
27	01:34:37	120	48	7	0.182	8	3,640	687,707	1,330	494	0	0	aborted_streaming
28	01:36:38	10	10	1	0.069	4	2,349	231,213	569	58	0	0	completed
29	01:36:53	75	17	3	0.062	4	1,706	233,562	332	111	0	0	aborted_streaming
30	01:38:08	30	25	2	0.127	6	2,701	472,872	944	181	0	0	completed
31	01:38:43	105	52	5	0.175	8	2,705	717,998	877	357	0	0	aborted_streaming
32	01:40:28	29	29	1	0.072	4	2,245	240,674	625	0	0	0	completed
33	01:40:58	22	22	2	0.129	6	2,598	487,593	918	190	0	0	completed
34	01:41:23	65	11	3	0.069	4	1,808	245,517	559	200	0	0	aborted_streaming
35	01:42:28	65	9	3	0.074	4	2,637	247,325	551	168	0	0	aborted_streaming
Σ		**3,750**	**1,498**	**156**	**6.10**	**242**	**324,577**	**14,199,873**	**66,643**	**15,211**	**0**	**6**	

Cost and API time: Claude Code reports total_cost_usd and duration_api_ms cumulatively across resumed sessions, so each row shows the increase over the previous session and Σ is the final cumulative value. Cost is Claude Code's own estimate, not an AWS bill. Wall s, turns and token counts are per session. Input tok = uncached input; cache write/read = prompt-cache creation/reads; output includes thinking. Terminal reason aborted_streaming = the session was cut off mid-response (consistent with CORAL interrupting and resuming the agent).



Agent actions

Tool	Calls
Bash	87
WebFetch	2
Write	1
Agent	1
SubagentHandback	1
WebSearch	1

coral command	Count
eval	21
diff	18
checkout	10
eval --tune	8
log	6
wait	2

Tool calls are those at the top level of the agent's transcript; the 'Bash' count includes every shell command, including file edits done via heredocs/sed. coral eval invocations (29) exceed recorded attempts (28) by one.

LLM gateway (LiteLLM → Bedrock)

Requests by model and HTTP status

Model alias	200	400	503	Total	Success %
claude-opus-5-5	187	38	2	227	82.4
claude-sonnet-5	0	29	0	29	0.0
all	**187**	**67**	**2**	**256**	**73.0**

Failure causes

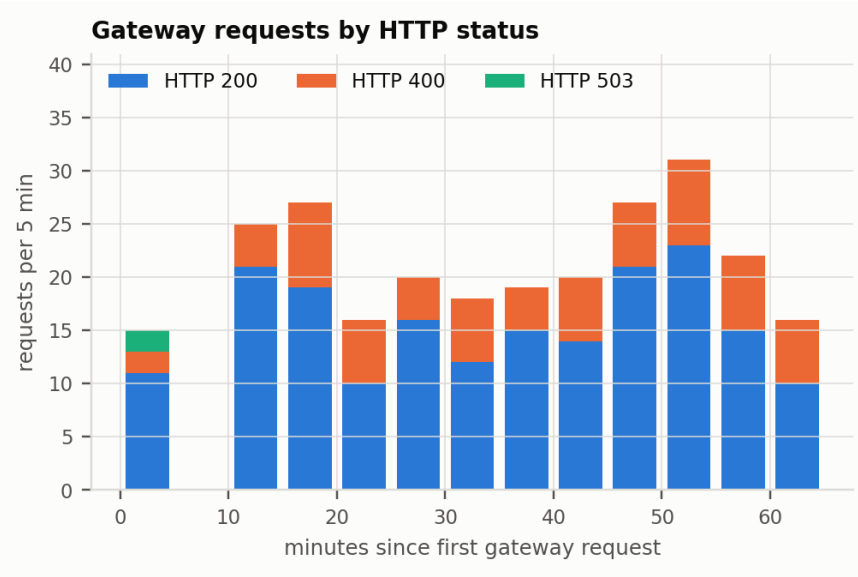
HTTP	Model alias	Count	Message (summarised from the gateway's error body)
400	claude-opus-5-5	36	messages.N.output_config: extra inputs not permitted
400	claude-sonnet-5	29	thinking.type.disabled not supported for this model
503	claude-opus-5-5	2	Bedrock unable to process request (503)
400	claude-opus-5-5	2	web_search tool type not supported

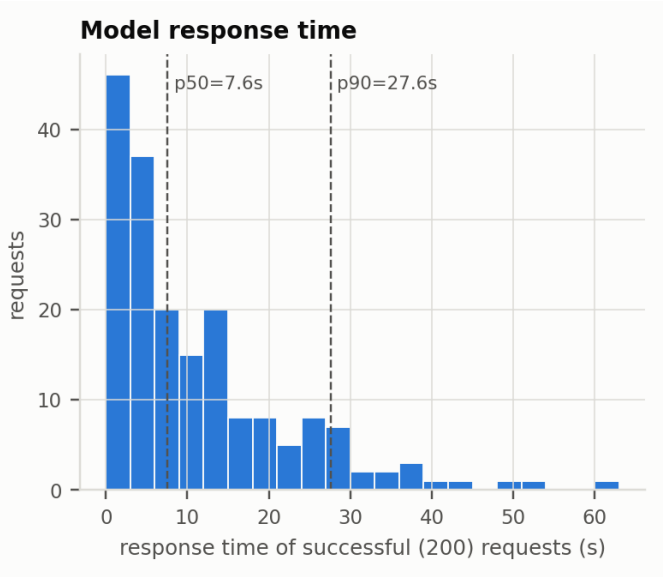
Response time of successful (200) requests

Model alias	n	Mean s	p50 s	p90 s	p99 s	Max s	Sum s
claude-opus-5-5	187	11.82	7.58	27.55	52.97	63.15	2,209

Totals over the gateway log

Metric	Value
Gateway requests logged	256
Distinct gateway sessions	29
Requests per gateway session (min / median / max)	1 / 7 / 42
Reported usage in 200 responses: input_tokens / output_tokens	54,141 / 72,226 (input excludes cached-prompt tokens)
Requests during the Bedrock 503 episode	2





Context the agent received and artifacts it wrote

Instruction files injected into the agent's prompts (Claude Code loads CLAUDE.md files up the directory tree)

File	Requests containing it	of total
~/arjua-forks/CLAUDE.md	252	256
~/arjua-forks/…/2026-10-09_004013/agents/captain-nemo/CLAUDE.md	252	256

The first path is the arjua-forks repository's own project instructions (the previous throughput-profiling project), because the agent's worktree sits below that directory. The second is the task's CLAUDE.md written by CORAL.

Notes written by the agent (.coral/public/notes)

File
_open-questions.md
_synthesis/s3fifo-scan-tradeoff.md
_synthesis/team-roster.md
experiments/eval-1-reference-get-scan.md
experiments/eval-10-ghost4x-gated.md
experiments/eval-11-small8-gated.md
experiments/eval-12-scan-ghost-main-head.md
experiments/eval-13-get-aware-aging-1.md
experiments/eval-14-get-aware-aging-2.md

File
experiments/eval-2-reference-s3fifo.md
experiments/eval-3-reference-mglru.md
experiments/eval-4-s3fifo-ignore-scan-accesses.md
experiments/eval-5-s3fifo-crp.md
experiments/eval-6-s3fifo-noise.md
experiments/eval-7-s3fifo-small10.md
experiments/eval-8-s3fifo-ghost2x.md
experiments/eval-9-ghost-gated-by-thread.md
focus/focus-get-aware-aging.md

Measurements not available for this run

Quantity	Status
Per-operation latency (READ p99, SCAN p99), per-thread throughput	Written by the benchmark to a per-eval scratch directory that is deleted after each eval; not retained
Policy counters (promotions, list sizes, ghost hits)	No policy exported a metrics file (policy_counters probe: 'no policy_metrics file')
CPU utilisation / cycles / cache misses during evals	Not collected
Noop baseline individual runs	Only the calibration mean and sd (n=8) were retained, not the 8 values
Per-eval timestamps of grading	Derived from attempt-file modification times (submit time = commit time)